SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct (192511387)
I Beffy. A. Shanika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: Beffy

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

1) media company - cloud Video streaming

Cloud Architecture:

* Use a content Delivery Network (CDN) to deliver videos from server closer to users.

* Store video files in cloud object storage.

* Deploy multiple Application servers behind a Load Balancer.

* Use Auto-scaling Groups to increase or decrease servers automatically.

* Use caching (Redis / memcached) to store frequently accessed content.

Load Balancing and Auto-Scaling

* Load balancing distributes user requests evenly across servers.

* It prevents server overload and improves performance.

* Auto - scaling automatically adds servers during peak traffic and removes them when demand decreases.

* This ensures smooth video streaming with minimal buffering.

Cost Optimization:

* use pay - as - you - Go pricing
purchase Reserved instances for regular workloads.

* use CDN caching to reduce bandwidth costs.

* monitor resources and remove unused instances

* optimize storage by moving old videos to low - cost storage.

* use spot instances for non- critical workloads to reduce computing costs significantly.

2) Fintech Startup - Multi - cloud Containers

Container Orchestration:

* Deploy applications using Kubernetes.
* Kubernetes manages deployment, scaling, and update automatically.
* containers provide consistency accross different cloud providers.

Service Discovery and Scaling:

* service discovery helps containers locate and communicate with each other automatically.
* Horizontal scaling increases container replicas during high demand
* Load balancing distributes traffic among container instances.

Benefits:

* High availability
* Better disaster recovery
* Vendor independence
* Improved scalability

Risks:

* complex management
* Higher operational cost
* security and networking challenges
* Difficult monitoring across multiple clouds.

3) E-Learning platform:

Backup and Disaster Recovery solution

* use cloud-native backup to automatically back up student records, course materials, and application data.

* Enable data replication across multiple cloud regions to prevent data loss if one region fails.

* Use object versioning so deleted or modified files can be restored to previous versions.

* Store backups in seperate locations to ensure data availability during disasters.

Recovery Time Objective (RTO) and Recovery Point objective (RPO)

* RTO (Recovery Time Objective): The maximum time allowed to restore services after a failure. This is achieved using automated recovery and failover mechanisms.

* RPO (Recovery Point objective): The maximum acceptable amount of data loss. Frequent backups and continuous replication help keep RPO very low.

* Regular testing of the disaster recovery plan ensures that RTO and RPO targets are met.

Business continuity:

* Performs regular backup testing
* Use automated failover systems
* Monitor backup status continuously
* Maintain a disaster recovery plan.

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4) Government Agency - Hybrid cloud Model

Hybrid cloud Architecture:

* Store sensitive data in the on-premises data center.
* Store non-sensitive data in the public cloud
* Connect both environment using a secure VPN or private network
* Use a hybrid cloud platform to manage both environments efficiently.

Identity Management and Encryption

* Implement identity and Access management (IAM)

* Use Role-Based Access control (RBAC) for authorized access

* Enable Multi-factor Authentication (MFA)

* Encrypt data at rest and in transit.

Secure Communication:

* Use HTTPS and SSL/TLS for secure data transfer

* Protect the network with firewalls.

* Monitors traffic to detect unauthorized access

* Perform regular security audits.

Challenges in Hybrid cloud:

* complex infrastructure management

* compliance with government regulations.

Integration between cloud and on-premises systems.

Network latency and security risks.

5) SaaS Provider - High Availability and Fault Tolerance

Multi - Region Deployment Strategy:

* Deploy the application in multiple cloud regions

* Use a Global Load Balancer to distribute user traffic across regions.

* Keep backup servers in another region to ensure service availability.

* If one region fails, traffic is automatically redirected to the healthy region.

Data Replication and Failover:

* Enable real-time data replication between regions.

* Store copies of data in multiple locations to prevent data loss.

* Configure automatic failover to switch users to the backup region during failures.

* Regularly test failover mechanisms to ensure they work properly.

Monitoring and Alerting:

* Monitor CPU, memory, storage, and network usage

* Configure alerts for failures and performance issues

* Use centralized logging and dashboard

* Respond quickly to incidents to maintain continuous service.

Fault Tolerance:

* Deploy multiple application instances to avoid a single point of failure

* Replace failed servers automatically using auto-scaling.

* Perform regular health checks to ensure all services are running properly

* Maintain backup systems for critical services.